Roll. No.: _____ Name: _____ Section: ____

CSE638: Graduate Systems
Midsem Winter 2025
Date: February 23, 2025; Duration: 60 mins (10 am to 11 am)
Total marks: 30

Instructions:

(1) Read the questions carefully and answer.
(2) **Answer to the point as much as possible.** Include the important technical details.
(3) If you make practical & appropriate assumptions, mention them clearly.

Q.1. "When an interrupt occurs, the process switches from user mode to kernel mode for interrupt processing. To start the execution of interrupt code, the system has to fetch the process image of OS into the memory" State True or False; No marks without justification. **[2]**

Q.2. Given that a regular process is currently running on the CPU. State at least FOUR distinct events (i.e., type of events) when the Operating System takes control of the system. **[4]**

Q.3. For each of the following variables declared in a C program, state which part of the memory image of the process will contain the memory allocated to the variable. **[4]**

(a) A global variable declared outside any function in the program.
(b) A static variable declared inside a function definition.
(c) An integer variable declared inside the main function.
(d) The variable "ptr" in the following line of code located inside the main function
int *ptr = malloc(sizeof(int));

Q.4. You are using a Virtual Machines (VM) for system virtualization. The application running inside a VM accesses the system resources such as CPU, memory, and I/O. Answer the following questions related to CPU virtualization across VMs. **[2+2+3=7]**

(a) Assume traditional trap-and-emulate mechanism is implemented (without hardware-assisted virtualization). Will the system work correctly under the following scenarios? If "yes" tell how, and if "no" tell why? **[2]**
  (i) CPU instruction "X" requires access to system resources.
  (ii) CPU instruction "Y" can run in both "priviliged" and "non-privileged" mode with a different outcome.
(b) Imagine a scenario where your VM always runs applications that execute **unprivileged instructions 99.9%** of the time, and **privileged instructions 0.1%** of the time. Would you prefer "paravirtualization" or "full virtualization" (note: you have no other choice)? Why? **[2]**
(c) Suppose QEMU/KVM architecture is used for hardware-assisted CPU virtualization. What are the major steps involved when a Guest application executes a privileged instruction? Include the data structures used to store context. **[3]**

Q.5. Consider an architecture that uses **N-level hierarchical paging**, where all page tables (traditional (or regular), extended, shadow, and so on) have **"N" levels**. A hypervisor on this system runs a **single guest VM**, which in turn has **TEN active processes**. **[4]**

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(a) If the hypervisor uses **extended page tables (EPT)** for memory virtualization
  (i) How many extra page tables does the hypervisor maintain to virtualize the guest, in addition to the page tables maintained by the guest VM itself?
  (ii) How many memory accesses (approximately) must the MMU perform in order to "walk the page table" and translate ONE address?

(b) If the hypervisor uses **Shadow page tables** for memory virtualization
  (i) Same Question as 5(a)(i)
  (ii) Same Question as 5(a)(ii)

Q.6. Suppose you are running a web application that communicates with the web server over the Internet over a 1 Gbps network cable. If you somehow know the following. **[3]**

(a) The bottleneck bandwidth from your machine to the web server is 10^6 bits/sec
(b) The time required to send your data is 1 millisecond (i.e., ms)
(c) The time requires to receive an ACK is 1 millisecond

What should be the TCP window size for the web application to maximise the network throughput?

Q.7. Complete the following code snippet by filling in the blanks: **[6 * 0.5 = 3]**

```
typedef struct {. . .} arg_t; // Dummy; Don't fill this
void *thread_func(void *arg) {
        . . . // Dummy; Don't fill this
}
int main(int argc, char *argv[]) {
    pthread_t p;
    arg_t args = { 10, 20 };

    printf("running 'thread_func' with 'args'\n");
    int rc = pthread_create(&p, _____, _____, _____);// (A)

    printf("waiting for 'thread_func'\n");
    __________(_____, _____); // (B)
    printf("done\n");
    return 0;
}
Ans. (A) ____, _____, _____ (B) ______(______, ______)
```

Q.8. What system calls will be used for the following execution sequence: **[3]**

(a) Process "P1" wants create a new process "P2" and place it into a new namespace "ns1"
(b) Process "Q1" is in the default namespace. Q1 wants to create a new process "Q2" and place it into namespace "ns1"
(c) Process "P2" wants to create a new process "P3" in namespace "ns1"

_____THE END_____